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Moreno, SD, Brown, LE, Coburn, JW, and Judelson, DA. Effect of cluster sets on plyometric jump power. *J Strength Cond Res* 28(9): 2424–2428, 2014—Cluster sets may lead to enhanced power (PW) production by allowing for partial recovery. The purpose of this study was to determine the effects of cluster sets vs. traditional sets on plyometric jump PW, ground reaction force, take-off velocity (TOV), and jump height (JH). Twenty-six recreationally trained college men completed 3 testing sessions, which involved performing repeated body-weight (BW) plyometric squat jumps across 3 different set configurations: traditional (2 sets of 10 with 90-second rest between sets), cluster 1 (4 sets of 5 with 30-second rest between sets), and cluster 2 (10 sets of 2 with 10-second rest between sets). Ground reaction force results demonstrated no interaction or main effect for condition, but there was a significant ($p \leq 0.05$) main effect for repetition, where repetition 1 was significantly less than repetitions 3–5, 7–10, 12–15, and 17–20. For TOV, PW, and JH, there were significant interactions. Take-off velocity resulted in the following: Traditional, repetition 1 was significantly greater than repetitions 7–10 and 17–20, but was significantly less than repetition 13; cluster 1, repetition 1 was significantly less than repetitions 2–5; and cluster 2, there were no significant differences. Power resulted in the following: Traditional, repetition 1 was significantly greater than repetitions 4–10 and 14–20; cluster 1, repetition 1 was significantly greater than repetitions 7–10 and 12–20; and cluster 2, repetition 1 was significantly greater than repetitions 3, 6–18, and 20. Jump height resulted in the following: Traditional, repetition 1 was significantly greater than repetitions 18–20, but was significantly less than repetitions 3 and 13. For cluster 1 and cluster 2, there were no significant differences. These results demonstrate that cluster sets, specifically 10 sets of 2, allow for a greater maintenance of PW, TOV, and JH compared with a traditional 2 sets

KEY WORDS speed, velocity, height, force

Plyometric training is commonly used to develop the efficiency of an athlete's stretch-shortening cycle (SSC) (16). The SSC is described as the successive combination of eccentric and concentric actions (4), which is used in most sport movements, such as throwing, running, and jumping (16). Efficient SSC mechanics can result in energy conservation and enhanced power (PW) output, ultimately, leading to improved performance (16,19). Knowing the importance of the SSC mechanism, it is imperative that an optimally designed plyometric program is administered by coaches.

It is common to see PW exercises, such as cleans and snatches, performed with sets of 3 repetitions or less. This may be because of the fact that bar velocity or displacement or both begin to decrease with a greater amount of repetitions, thus reducing PW, and ultimately defeating the purpose of training for PW (2,9). Baker and Newton (2) found progressively decreasing PW outputs after the fifth repetition when performing 60-kg bench throws or 60-kg squat jumps. Haff et al. (9) showed that velocity and displacement of a barbell decreased with each consecutive repetition when performing clean pulls. Also, Gorostiaga et al. (7) noticed a significant drop in PW output during the last 5 repetitions of a 10 repetition maximum (RM) leg press. These studies all support performing limited repetitions of an exercise if PW is the goal.

In attempt to maximize PW output, Haff et al. (8,9) offered a nontraditional resistance set configuration, called cluster training, involving the use of inter-repetition or intraset rest intervals. Cluster sets are theorized to allow for some replenishment of phosphocreatine (PCr) (8), which is vital for strength and PW production (12,17,18). Haff et al. (8,9) proposed that performance characteristics, such as peak PW output, barbell velocity, and displacement, will decrease with

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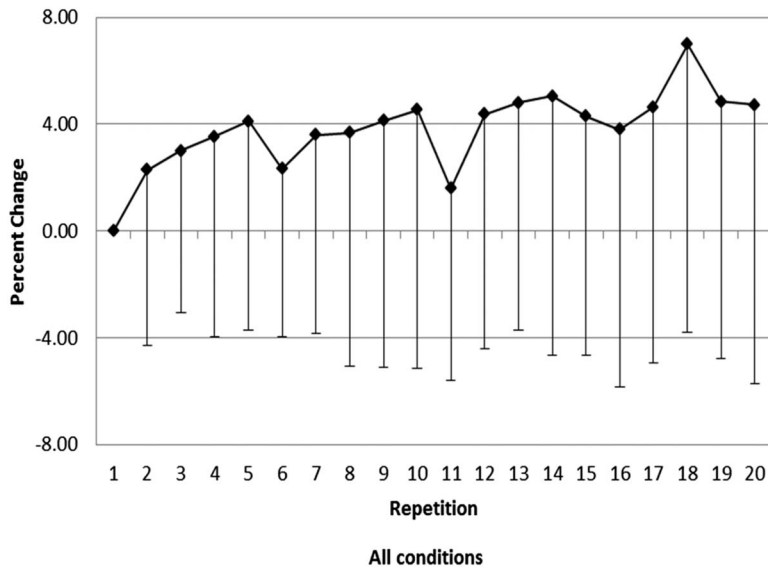


Figure 1. Peak ground reaction force (mean and SD) across the 20 repetitions for all conditions combined. Results are presented as percent change from the average of the first repetition of all 3 conditions. Repetition 1 was significantly less than repetitions 3–5, 7–10, 12–15, and 17–20.

each subsequent repetition in a traditional set with no inter-repetition rest. Also, Baker and Newton (1) suggest that it is important to avoid fatigue when attempting to maximize PW output, and that performing a low amount of repetitions, with an appropriate rest interval, will optimize PW training.

ric training should include a limited number of consecutive repetitions, similar to explosive barbell exercises, to maximize PW. Therefore, the purpose of this study was to assess the effects of cluster sets on plyometric PW. The hypothesis was that cluster sets will result in greater plyometric PW than a traditional plyometric set configuration.

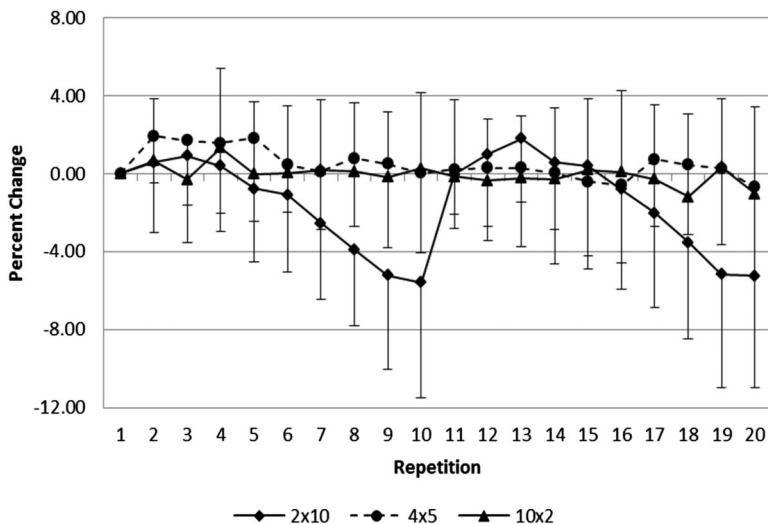


Figure 2. Peak take-off velocity (mean and SD) across the 20 repetitions by condition. Results are presented as percent change from the first repetition of each condition. For the traditional 2×10 sets, repetition 1 was significantly greater than repetitions 7–10 and 17–20, but was significantly less than repetition 13. For the 4×5 cluster sets, repetition 1 was significantly less than repetitions 2–5. For the 10×2 cluster sets, there were no significant differences.

Plyometrics and explosive barbell exercises are different methods of training, yet both are used to increase PW output. With plyometric training, there seems to be some discrepancy in the optimal number of consecutive repetitions performed (3). The reason may be that there are so many different plyometric exercises, along with many variations of intensity.

The current knowledge on cluster sets suggests that they allow for greater PW outputs when compared with traditional set configurations (6,8–11,13). It is suggested that cluster sets may be best suited for explosive exercises (8). However, there seems to be a lack of information on the effects of cluster sets on plyometric exercises. Yet, it seems plausible that plyomet-

METHODS

Experimental Approach to the Problem

This was a within-subjects study design that examined the effects of cluster sets on repeated plyometric jumps. The set configurations consisted of traditional with 2 sets of 10 jumps with 90-second rest between sets, cluster 1 with 4 sets of 5 jumps with 30-second rest between sets, and cluster 2 with 10 sets of 2 jumps with 10-second rest between sets. The dependent variables measured were ground reaction force (GRF), take-off velocity (TOV), PW, and jump height (JH). In this model, jump time and rest

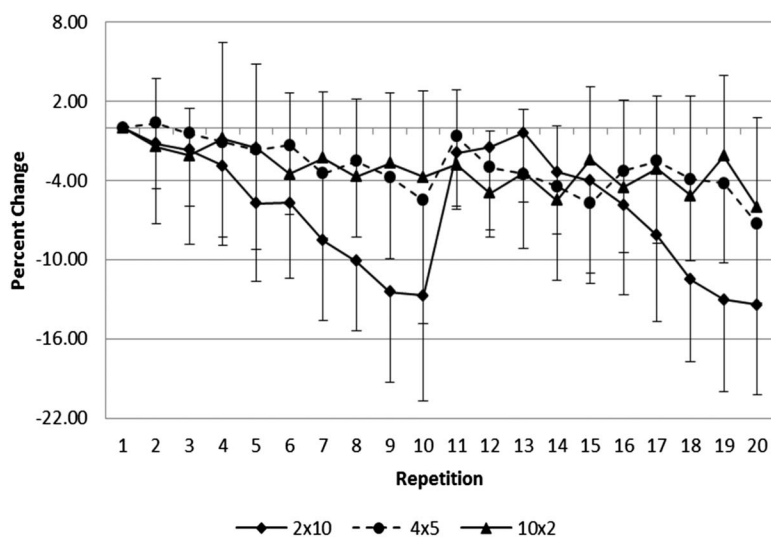


Figure 3. Peak power (mean and SD) across the 20 repetitions by condition. Results are presented as percent change from the first repetition of each condition. For the traditional 2×10 sets, repetition 1 was significantly greater than repetitions 4–10 and 14–20. For the 4×5 cluster sets, repetition 1 was significantly greater than repetitions 7–10 and 12–20. For the 10×2 cluster sets, repetition 1 was significantly greater than repetitions 3, 6–18, and 20.

times were manipulated so that total exercise time was the same for all 3 conditions. This was performed to attempt to determine optimal configuration for performance (2,7–9).

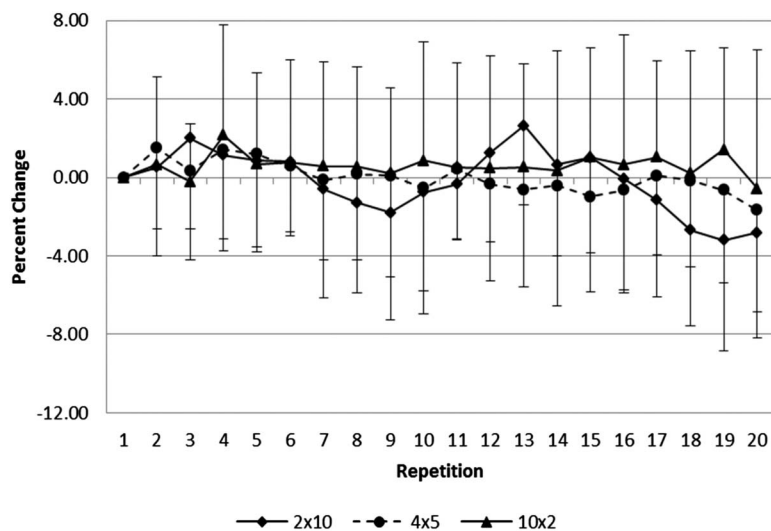


Figure 4. Jump height (mean and SD) across the 20 repetitions by condition. Results are presented as percent change from the first repetition of each condition. For the traditional 2×10 sets, repetition 1 was significantly greater than repetitions 18–20, but was significantly less than repetitions 3 and 13. For the 4×5 and 10×2 cluster sets, there were no significant differences.

Subjects

Twenty-six healthy men (age = 22.32 ± 2.1 years; age range 19–28; height = 178.11 ± 5.89 cm; mass = 81.00 ± 9.53 kg) with no history of lower-body injury in the past 6 months volunteered to participate. Subjects must have been participating in some type of jumping and lower-body resistance exercise at least 1 day a week over the past 6 months. All subjects were instructed to maintain their normal eating and sleeping habits throughout the study. They were also encouraged to fully hydrate before each testing session which was held at the same time of the day for each condition. Before participation, each subject provided written informed consent approved by the University Institutional Review Board.

Procedures

Subjects performed repeated body-weight (BW) plyometric jumps with their hands on their hips in 3 different plyometric set configurations, which included traditional set (2), consisting of 2 sets of 10 jumps separated by 90-second rest between sets, cluster set 1, consisting of 4 sets of 5 jumps with 30-second rest between sets, and cluster set 2, consisting of 10 sets of 2 jumps with 10-second rest between sets. Therefore, each condition had 20 jumps but rest time between sets was manipulated so every condition took 2 minutes of the total time. Participation involved 3 visits separated by 7 days with a different random condition on each visit. Each session took approximately 15 minutes.

Visit 1 consisted of the informed consent, measurements of height and mass, instructions for the study, and a dynamic warm-up consisting of 2 sets of 10 BW squats, 10 BW lunges, 10 high-kicks, 1 set

of 5 BW squat jumps, then a few practice jumps. This was followed by 2-minute rest then by one of the 3 set configurations in random order.

Visits 2 and 3 consisted of the same dynamic warm-up, 2-minute rest, and one of the other set configurations. Subjects were told to refrain from any strenuous lower-body activities, such as resistance training, running, and sport participation for 48 hours before each visit.

An AMTI force plate (Advanced Mechanical Technology, Inc., Watertown, MA, USA) was used to measure GRF. Data were sampled at 1,000 Hz, and the force plate was connected to a desktop computer running custom LabVIEW data collection and analysis software (version 7.1, National Instruments Corporation, Austin, TX, USA) to analyze the force-time and velocity-time curves, which were used to determine PW (force $\times$ velocity) and JH (time in the air equation). To reduce the error of measurement, subjects were instructed to try jump and land in the same spot for each repetition. Reliability analysis showed that intraclass correlation coefficient values ranged between 0.92 and 0.97.

Statistical Analyses

The Statistical Package for the Social Sciences (SPSS, version 19.0; SPSS, Inc., Chicago, IL, USA) was used for all data analyses. Alpha was set a priori at 0.05. Four 3×20 (condition $\times$ repetition) repeated-measure analyses of variance (ANOVAs) were used to evaluate each dependent variable of GRF, TOV, PW, and JH. Follow-up simple ANOVAs were used to evaluate interactions and main effects.

RESULTS

For GRF, there was no interaction or main effect for condition, but there was a main effect for repetition. A 1×20 post hoc ANOVA revealed that repetition 1 was significantly less than repetitions 3–5, 7–10, 12–15, and 17–20 (Figure 1).

For TOV, there was a significant interaction. This was followed up with three 1×20 ANOVAs, 1 for each condition. For the traditional 2×10 sets, repetition 1 was significantly greater than repetitions 7–10 and 17–20, but was significantly less than repetition 13. For the 4×5 cluster sets, repetition 1 was significantly less than repetitions 2–5. For the 10×2 cluster sets, there were no significant differences (Figure 2).

For PW, there was a significant interaction. This was followed up with three 1×20 ANOVAs, 1 for each condition. For the traditional 2×10 sets, repetition 1 was significantly greater than repetitions 4–10 and 14–20. For the 4×5 cluster sets, repetition 1 was significantly greater than repetitions 7–10 and 12–20. For the 10×2 cluster sets, repetition 1 was significantly greater than repetitions 3, 6–18, and 20 (Figure 3).

For JH, there was a significant interaction. This was followed up with three 1×20 ANOVAs, 1 for each condition. For the traditional 2×10 sets, repetition 1 was significantly

greater than repetitions 18–20, but was significantly less than repetitions 3 and 13. For the 4×5 and 10×2 cluster sets, there were no significant differences (Figure 4).

DISCUSSION

This study demonstrated that cluster sets resulted in a greater maintenance of plyometric jump PW, whereas the traditional sets allowed for near-complete recovery between sets. Many prior studies performed on cluster sets were performed on strength exercises or explosive barbell exercises (6,9–11,13) and many studies compared cluster sets with a traditional set without controlling rest time, resulting in disparate total exercise times (9,11,13). In this study, total rest time and total volume were the same in all conditions, resulting in equal total workout times. This is important because many strength and conditioning professionals have a fixed amount of time to train their athletes, thus making our protocol more practical for coaches. It is also important to note that our results compare all repetitions to the first repetition of each condition and that only peaks of the variables were measured.

Most previous research supports cluster sets for barbell training (9–11,13). Haff et al. (9) showed that cluster sets allowed for higher peak velocity and displacement than a traditional set when performing clean pulls with 90 and 120% of one's 1RM clean. Lawton et al. (13) showed greater total PW outputs in a 6RM bench press when performing cluster sets compared with a traditional set. Hansen et al. (10) demonstrated greater peak PW and peak velocities, among cluster sets, for many of the latter repetitions when performing multiple sets of jump squats, when compared with a traditional set. Also, Hardee et al. (11) showed that inter-repetition rest allowed for greater maintenance of PW and velocity than a traditional set format in the PW clean.

In this study, PW tended to decrease gradually with succeeding repetitions in most sets for all conditions, but was markedly noticeable after the third repetition of each set in the 2×10 condition. This is in agreement with Baker and Newton (2) who demonstrated decreased PW outputs after the fifth repetition in a traditional set, whereas peak PW was seen in the second and third repetitions of jump squats and bench throws, respectively. Furthermore, Gorostiaga et al. (7) showed decreases in PW output after the first 5 repetitions of a 10RM leg press, which was associated with a decreased contribution of PCr and an increase in muscle lactate. This could explain the lower PW output seen in the later repetitions of the 2×10 condition, as subjects decreased the use of the PCr system. Interestingly, PW was maintained for the first 5 jumps of the 4×5 condition as opposed to only the first 3 jumps of the 2×10 condition. This may be explained by a psychological component in that the subjects tried to conserve energy and pace themselves in the 2×10 condition because of the extended number of repetitions. Overall, most studies have shown that cluster sets attenuate decreases in PW, which was attributed to PCr resynthesis and removal of lactate (6,10,11,13).

Another important observation in this study was that the 2×10 condition had comparable sets, which started with high PW outputs, but then gradually declined after the third repetition of each set. The long interset rest interval of the 2×10 condition allowed for near-complete recovery, which was expressed by the higher PW outputs in the initial repetitions of the second set. However, the 4×5 and 10×2 conditions showed lower initial PW outputs in most sets after the first and third sets, respectively, but never fell to that of the last repetitions of the 2×10 condition. In this study, the 2×10 condition had a greater work to rest ratio at 1:9, in comparison to the 4×5 and 10×2 conditions, which had 1:6 and 1:5 ratios, respectively. It has been demonstrated that the greater the amount of PCr degradation, the greater the rest required for full PCr replenishment (5,14,15). The lower initial PW outputs of the 4×5 and 10×2 conditions may be explained by the smaller work to rest ratio, which did not allow for maximal recovery as seen in the 2×10 condition.

There were no differences between groups in GRF, but there was a significant repetition main effect, which consisted of a steady increase in GRF across most repetitions. These results were puzzling, and it was difficult to find any prior research to support our findings. The slight increase in force may be a function of the eccentric loading coming from the descent. Perhaps, increasing fatigue in the leg muscles, as repetitions proceeded, led to a decrease in their control of the countermovement, causing a more rapid descent and ultimately greater GRF. Increases in GRF were coupled with no change or a slight decrease in velocity, leading to decreases in JH for, solely, the last 3 repetitions of the 2×10 condition. On the contrary, JH was significantly higher in the 2×10 condition for repetitions 3 and 13, which may have resulted from the great eccentric forces and low level of fatigue (2).

PRACTICAL APPLICATIONS

This study demonstrated that cluster sets helped maintain plyometric squat jump PW output, but did not allow for maximal recovery. The 2×10 condition had the greatest work to rest ratio, which allowed for maximal recovery between sets. However, there were substantial decreases in PW after the third and fifth repetitions in the 2×10 and 4×5 conditions, respectively. Based on these results, one should execute no more than 5 repetitions to avoid decreases in PW output and use a 1:9 work to rest ratio to allow for maximal recovery. Also, athletes might perform more than 2 repetitions to elicit the greater eccentric loads derived from successive jumps. Although this was not a training study, overall our data suggest that coaches have their athletes perform 2–5 jumps with 27–45 seconds of rest, which will attenuate fatigue and allow for greater eccentric loads of successive repetitions. Future research should examine varied rest times

between cluster sets while simultaneously measuring blood lactate in an attempt to elucidate physiological mechanisms associated with performance decrements.

REFERENCES

1. Baker, D and Newton, R. Methods to increase the effectiveness of maximal power training for the upper body. *J Strength Cond Res* 27: 24–32, 2005.
2. Baker, D and Newton, R. Change in power output across a high-repetition set of bench throws and jump squats in highly trained athletes. *J Strength Cond Res* 21: 1001–1011, 2007.
3. Chu, DA. *Jumping Into Plyometrics*. Champaign, IL: Human Kinetics, 1998.
4. Cormie, P, McGuigan, MR, and Newton, R. Developing maximal neuromuscular power: Part I. *J Sports Med* 41: 17–38, 2011.
5. Dawson, B, Goodman, C, Lawrence, S, Preen, D, Polglaze, T, Fitzsomons, M, and Fournier, P. Muscle phosphocreatine repletion following single and repeated short sprint efforts. *Scand J Med Sci Sports* 7: 206–213, 1997.
6. Denton, J and Cronin, JB. Kinematics, kinetic, and blood lactate profiles of continuous and intraset rest loading schemes. *J Strength Cond Res* 20: 528–534, 2006.
7. Gorostiaga, E, Navarro-Amézqueta, I, Cusso, R, Hellsten, Y, Calbet, J, Guerrero, M, Granados, C, González-Izal, M, Ibáñez, J, and Izquierdo, M. Anaerobic energy expenditure and mechanical efficiency during exhaustive leg press exercise. *PLoS One* 5: e13486, 2010.
8. Haff, GG, Hobbs, RT, Haff, EE, Sands, WA, Pierce, KC, and Stone, MH. Cluster training: A novel method for introducing training program variation. *J Strength Cond Res* 30: 67–76, 2008.
9. Haff, GG, Whitley, A, McCoy, LB, O'Bryant, HS, Kilgore, JL, Haff, EE, Pierce, K, and Stone, MH. Effects of different set configurations on barbell velocity and displacement during a clean pull. *J Strength Cond Res* 17: 95–103, 2003.
10. Hansen, KT, Cronin, JB, and Newton, MJ. The effect of cluster loading on force, velocity, and power during ballistic jump squat training. *Int J Sports Physiol Perform* 6: 455–468, 2011.
11. Hardee, JP, Triplett, NT, Utter, AC, Zwetsloot, KA, and McBride, JF. Effect of interrepetition rest on power output in the power clean. *J Strength Cond Res* 26: 883–889, 2012.
12. Kraemer, WJ and Ratamess, NA. Fundamentals of resistance training: Progression and exercise prescription. *Med Sci Sports Exerc* 36: 674–688, 2004.
13. Lawton, TW, Cronin, JB, and Lindsell, RP. Effect of interrepetition rest intervals on weight training repetition power output. *J Strength Cond Res* 20: 172–176, 2006.
14. Read, MM and Cisar, C. The influence of varied rest interval lengths on depth jump performance. *J Strength Cond Res* 15: 279–283, 2001.
15. Spencer, M, Bishop, D, Dawson, B, and Goodman, C. Physiological and metabolic responses of repeated-sprint activities specific to field-based team sports. *Int J Sports Med* 32: 1025–1044, 2005.
16. Turner, AN and Jeffreys, I. The stretch-shortening cycle: Proposed mechanisms and methods for enhancement. *J Strength Cond Res* 32: 87–99, 2010.
17. Weiss, LW. The obtuse nature of muscular strength: The contribution of rest to its development and expression. *J Appl Sports Sci Res* 5: 219–227, 1991.
18. Willardson, JM. A brief review: How much rest between sets? *J Strength Cond Res* 30: 44–50, 2008.
19. Zatsiorsky, VM and Kraemer, WJ. *Science and Practice of Strength Training*. Champaign, IL: Human Kinetics, 2006.